



Department of Computer Science and Engineering

B.Tech. Computer Science and Engineering

Course Name : Responsible and Explainable AI
Course Code : TCS349
Semester : III
Pre-requisite : Fundamentals of AI and ML.

L	T	P	Credits
3	0	0	3

Course Learning Objectives:

CLO ₀₁	Identify the fundamental ethical dilemmas in AI and classify data, algorithmic, and interactional biases in AI systems.
CLO ₀₂	Describe tools for achieving AI transparency and apply accountability frameworks to balance security and privacy in AI implementations.
CLO ₀₃	Examine the long-term impact of AI on employment, economy, and public policy through case-based discussion.
CLO ₀₄	Critically assess the strengths and limitations of feature importance, perturbation analysis, and rule-based explanation techniques.
CLO ₀₅	Construct heatmap and partial-dependence-plot visualizations and propose practical XAI applications for finance, healthcare, or autonomous systems.

Course Outcomes:

After completion of the course, the students will be able to:

CO ₀₁	Explain the fundamental ethical principles, responsible AI practices, and societal impacts associated with the development and deployment of Artificial Intelligence systems.
CO ₀₂	Apply appropriate techniques to detect, measure, and mitigate bias in AI models for engineering applications while ensuring fairness and accountability.
CO ₀₃	Analyze transparency, accountability, privacy, and governance mechanisms to evaluate responsible AI systems in real-world scenarios
CO ₀₄	Evaluate Explainable Artificial Intelligence (XAI) methods, including local and global explanation techniques, for interpreting machine learning models across different application domains.
CO ₀₅	Design responsible and explainable AI solutions by integrating ethical principles, fairness, transparency, accountability, and explainability into the AI development lifecycle for engineering applications.

Details of the Course:



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HILL UNIVERSITY

Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

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Unit 1: **Contact Hours/Teaching Hours: 9**

UNIT:1 Introduction to AI Ethics, Bias in AI Systems, Fairness in AI

Fundamentals of ethics in AI, Importance of ethical AI development, Overview of common ethical dilemmas in AI, Types of biases in AI (data, algorithmic, interactional), Detecting and measuring bias, Mitigation strategies for reducing bias, Defining fairness in AI contexts, Fairness metrics and their trade-offs- Designing fair AI systems, Imminent risks from AI Models, Long-term risks from AI Models

Self-Learning Hours = 4

Students shall understand the concepts of ethics and biasness in AI.

Unit 2: **Contact Hours/Teaching Hours: 9**

Transparency and Accountability, AI in Surveillance and Privacy

Principles of RAI, Importance of transparency in AI, Tools and methods for achieving transparency, Accountability frameworks and regulatory compliance, AI applications in surveillance technologies, Privacy concerns and data protection laws, Balancing security and privacy in AI implementations

Self-Learning Hours = 4

Students shall use the various tools used in industry for AI.

Unit 3: **Contact Hours/Teaching Hours: 7**

Societal Impacts of AI

AI's impact on employment and economy, AI in governance and public policy, Long-term implications of AI on society.

Unit 4: Searching and Sorting

Contact Hours/Teaching Hours: 9

Explainable Artificial Intelligence (XAI) Fundamentals

Introduction to Explainable AI, Need and importance of explainability in AI, Overview of XAI methods, Local Explanation Techniques, LIME, SHAP, Global Explanation Techniques, Feature importance, Decision trees as explanatory models- Rule-based explanations.

Self-Learning Hours = 6

Student shall implement the LIME, SHAP techniques.

Unit 5: **Contact Hours/Teaching Hours: 08**

Visualization and Applications of Explainable AI

Visualization of feature effects and model decisions, Use of heatmaps and partial dependence plots- Network and layer activation visualizations in deep models



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XAI in finance (credit scoring, fraud detection), XAI in healthcare (diagnostic systems, treatment recommendations), XAI in autonomous systems (vehicle control systems, robotics)

Self-Learning Hours = 6

Students shall implement various visualization methods using python.

Team Activity Hours = 2

Group activity & discussions to provide solution of the real-world problems like fraud detection, disease detection etc .

Course Articulation Matrix

COs / POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	-	-	-	3	-	2	-	1	-	2	1	2	3
CO2	2	3	2	2	3	2	1	3	-	-	-	2	2	3	2
CO3	-	3	2	3	2	3	2	3	-	-	-	2	2	2	3
CO4	2	3	-	3	3	2	-	2	-	-	-	-	2	3	3
CO5	2	-	3	2	3	2	-	3	-	-	-	3	3	3	3

High correlation (3); Medium correlation (2); Low correlation (1), - No correlation

Text Books:

Authors Name	Title	Edition	Publisher, Country	Year
Stuart Russell, Peter Norvig	Artificial Intelligence: A Modern Approach	3rd Edition	Pearson, USA	2009
Michael Munn, David Pitman	Explainable AI for Practitioners: Designing and Implementing Explainable ML Solutions	1st Edition	O'Reilly, India	2022
John Danaher	Ethics of Artificial Intelligence	2nd Edition	Oxford University Press, USA	2022



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Reference Books:

Authors Name	Title	Edition	Publisher, Country	Year
Denis Rothman	Hands-On Explainable AI (XAI) with Python	1 st Edition	Packt Publishing, USA	2020
James Manyika, Michael Chui, and Michael Osborne	The Ethics of Artificial Intelligence	1 st Edition	McKinsey Global Institute, USA	2018